Product Inventory Management System

1. Introduction

A Product Inventory Management System is a software application used to track, manage, and organize products in a store or company. It helps businesses monitor stock levels, sales, purchases, and product details efficiently.

Inventory management ensures that the right products are available at the right time without overstocking or running out of items.

Team ID : SWTID-2026-6470

Team Size : 5

Team Leader : MANJARI T

Team member : MEGALA M

Team member : NITHYA SREE S

Team member : PRATHISHA S

Team member : RAMASRI D

2. Objectives

- To maintain accurate records of products.
- To track the quantity of items available in stock.
- To manage product purchases and sales.
- To reduce manual work and human errors.
- To improve business efficiency and decision making.

3. Features of Inventory Management System

- Product Management – Add, update, and delete product details.
- Stock Tracking – Monitor the quantity of each product in stock.
- Sales Management – Record product sales and update stock automatically.

- Purchase Management – Manage purchases from suppliers.
- Report Generation – Generate reports for stock, sales, and purchases.

4. Components of Inventory System

4.1 Product

Stores product information such as:

- Product ID
- Product Name
- Category
- Price
- Quantity

4.2 Supplier

Information about the suppliers who provide products.

4.3 Stock

Shows the available quantity of each product in the inventory.

4.4 Customer

Stores customer information when products are sold.

5. Advantages

- Saves time and effort.
- Reduces human errors.
- Helps in better stock control.
- Improves business productivity.
- Provides accurate reports.

6. Disadvantages

- Initial setup cost.
- Requires technical knowledge.
- System failure may affect operations.

7. Applications

- Inventory management systems are used in:
- Retail stores
- Supermarkets
- Warehouses
- Manufacturing industries
- E-commerce companies

Inventory product management

Ideation Phase

Brainstorm & Idea Prioritization Template

Brainstorming is used to generate creative ideas and identify the most suitable solution for a real-world problem. During the ideation phase, team members collaborated to discuss common issues faced in managing product inventories and explored multiple web-based solutions using the MERN stack. Ideas were evaluated based on feasibility, scalability, security, and learning outcomes.

Step-1: Team Gathering, Collaboration and

Select the Problem Statement

The team conducted a discussion session to identify common problems that could be solved using full-stack development. Several ideas were proposed such as:

- E-commerce product tracker
- Expense manager
- Task management system
- Cryptocurrency dashboard
- Inventory management system

After evaluating complexity, real-world usefulness, and alignment with MERN stack learning objectives, the team finalized the following problem statement:

Problem Statement:

Small businesses and individual users lack a simple, secure, and centralized system to Manage their product inventory. Many existing solutions are complex or require paid subscriptions. Therefore, the team decided to build a **JWT-based Product Inventory Management System** that allows users to securely manage their own products.

Step-2: Brainstorm, Idea Listing and Grouping

During brainstorming, all team members contributed ideas without restrictions. The ideas were then grouped into categories:

Authentication & Security Ideas

- JWT-based login system
- User-specific product access
- Secure password hashing
- Role-based access (future scope)

Core Inventory Features

- Add product
- Update product details
- Delete product
- View all products
- Search products by name

User Experience Ideas

- Simple dashboard UI
- Responsive design
- Clean product cards layout
- Fast loading using React

Technical Ideas

- MERN stack architecture
- REST API development
- MongoDB database design
- Middleware for route protection

After grouping, the team focused on ideas that were realistic within the project timeline.

Step-3: Idea Prioritization

The team prioritized ideas based on:

- Feasibility within deadline
- Learning value
- Practical real-world use
- Technical complexity

High Priority (Implemented Features)

- JWT Authentication (Register/Login)
- Private product CRUD operations
- MongoDB database integration
- Secure API routes
- Search functionality

Medium Priority (Enhancements)

- Product filtering
- Improved UI components
- Error handling improvements

Low Priority / Future Scope

- Role-based admin system
- Analytics dashboard
- Notifications for low stock
- Deployment and cloud hosting

Final Decision:

The team finalized the **Product Inventory Management System** as it provides strong learning
opportunities in authentication, backend API design, and frontend integration using MERN stack
technologies

Product Inventory Management System
Ideation Phase

Customer Problem Statement Template:

The Customer Problem Statement helps identify the challenges faced by users while managing
product inventories. Understanding the user's perspective allows the team to design a system
that is secure, simple, and efficient. By analyzing user needs, the project focuses on creating a

I am	Describe customer with 3-4 key characteristics - *who are they?*	Describe the customer and their attributes here
I'm trying to	*List their outcome or "job" the care about - what are they trying to achieve?*	List the thing they are trying to achieve here
but	Describe what problems or barriers stand in the way – *what bothers them most?*	Describe the problems or barriers that get in the way here
because	Enter the "root cause" of why the problem or barrier exists – *what needs to be solved?*	Describe the reason the problems or barriers exist
which makes me feel	Describe the emotions from the customer's point of view – *how does it impact them emotionally?*	Describe the emotions the result from experiencing the problems or barriers

I am	I'm trying to	But	Because	Which makes me feel
A small business owner	Manage and update product stock easily	Most inventory tools are complex or expensive	I need a simple system to quickly add and edit products	Frustrated managing inventory manually

Problem Statement	I am (Customer)	I'm trying to	But	Because	Which makes me feel
PS-1	A small business owner	Manage product stock efficiently	Existing tools are complicated or expensive	I need a simple system to add and update products quickly	Frustrated managing inventory manually
PS-2	A freelancer selling products online	Track available items and pricing	Data gets lost without a centralized platform	I want a secure dashboard where only I can manage my products	Stressed about losing important product data
PS-3	A shop manager	Update stock quantity regularly	There is no easy real-time CRUD system	I need a fast and responsive web app to update inventory anytime	Anxious when stock information is outdated
PS-4	A new entrepreneur	Organize products in one place	Many inventory apps require complex setup	I want a lightweight and user-friendly inventory solution	Overwhelmed by complicated software

8. Conclusion

A Product Inventory Management System is essential for businesses to manage their stock effectively. It helps in tracking products, reducing errors, and improving overall efficiency. Implementing such a system can greatly enhance business operations and customer satisfaction.